

ChatWala?au Demo Corpus

ChatWala?au is an open-source AI agent runtime distributed via PyPI as the ``chatwalaau`` package. The name combines ``chat`` with the Hawaiian word ``wala?au``, meaning ``to chat``, ``to talk``, or ``to converse``.

The runtime serves a single-page application that connects to a FastAPI backend. The backend orchestrates AI agents using Microsoft Agent Framework (MAF). Streaming chat is driven by the AG-UI protocol over Server-Sent Events.

Features include multimodal chat, voice input via Azure OpenAI Whisper or the new Realtime API, on-demand text-to-speech via ElevenLabs or Azure OpenAI Realtime voice models, image generation via Azure OpenAI Images, and Retrieval-Augmented Generation backed by a local ChromaDB store.

Demo Mode (Chapter 2)

Demo Mode is enabled by setting `DEMO_MODE=true` in the environment. When enabled, every metered external provider routes through an in-process dummy implementation:

- The chat client is replaced by `DemoChatClient`, which streams scripted scenarios from a keyword-routed lookup table.
- The STT provider returns a fixed bilingual sentence.
- The TTS provider returns a bundled placeholder MP3.
- `generate_image` and `edit_image` return one of two bundled PNG cards.
- The RAG embedder is replaced by a deterministic `stdlib` hash function that produces 1536-dimensional unit vectors matching the dimensionality of `text-embedding-3-small`.

Weather queries continue to use Open-Meteo because it is free and requires no API key. This is the one live external integration retained in demo mode so the demo has a real-feeling anchor.

Deployment Topology (Chapter 3)

The reference cloud target for ChatWala?au Demo Mode is Azure Web App for Containers. The Dockerfile under assets/docker/ produces a single image that bundles the React SPA inside the Python wheel and exposes port 8000. TLS terminates at the platform front door, so the application itself runs HTTP.

Authentication is NOT relaxed in demo mode. A demo bound to a non-loopback host MUST set either API_KEY (Bearer auth for CLI clients) or both AUTH_USERNAME and AUTH_PASSWORD_HASH (ID/PW + opaque session cookie for SPA users). The startup validators that warn on missing credentials fire identically in demo and live modes.

State resets on every container restart because the demo deployment does not mount Azure Files. The bundled demo corpus PDF is auto-ingested into ChromaDB during the FastAPI lifespan startup whenever the collection is empty.